TReqs Technologies

AI-native, Git-integrated, docs-as-code requirements management

Reqs-as-code

Git-integrated requirements management

```
<treqs-element id="1542faf2010311f1b22007581c079037" type="functional-requirement">

### REQ-001: Parse Numeric Consumption Array

The data loader shall accept an array of numeric values representing electricity consumption measurements
in kWh.

- Input: List of int or float values
- Output: Parsed numeric array
- Validation: All elements must be numeric types

<treqs-link type="addresses" target="1525e0de010311f1bc51bd180a14ab2b"/>

</treqs-element>
```

With TReqs, requirements are written in markdown (.md-files) and annotated using "treqs-elements". TReqs-elements can link to each other using "treqs-links", to create traceability between elements.

```
# <treqs-element id="3717796e010311f1bf5ef7eb82d97afb" type="implementation">
# parse_consumption_values: Parses array of numeric consumption values
# <treqs-link type="implements" target="1542faf2010311f1b22007581c079037"/>
# </treqs-element>

def parse_consumption_values(raw_values: list) -> list[float]:
    result = []
    for i, value in enumerate(raw_values):
        ...
```

TReqs-elements can be used to annotate anything text, like code implementation or unit tests. The code can then be traced back to the requirements, to give information about their relation

Metamodels, CLI & TReqs Check

```yaml
# Granular functional requirements derived from user stories
  # Implementation is optional (not all requirements need code)
  - name: functional-requirement
    outlinks:
      - type: addresses
        target: user-story
      - type: hasParent
        target: functional-requirement
      - type: relatesTo
    inlinks:
      - type: implements
        source: implementation
        required: 'false'
      - type: verifies
        source: unittest
        required: 'false'

  # Code implementations that fulfill functional requirements
  - name: implementation
    outlinks:
      - type: implements
        target: functional-requirement
      - type: hasParent
        target: implementation
      - type: relatesTo
    inlinks:
      - type: tests
        source: unittest
        required: 'true'

  # Unit tests that verify implementations
  - name: unittest
    outlinks:
      - type: tests
        target: implementation
      - type: verifies
        target: functional-requirement
      - type: hasParent
        target: unittest
```

```
philip@DESKTOP-PQ991MU:~/repos/demos/demo-electricity_app-be$ treqs check
```

UID	TYPE	LABEL	MESSAGE	FILE:LINE
4bae98ce027211f18a3fa9e079aec5e3	implementation	# ConsumptionRequest: Pydantic request model for API input validation	Required inlinks missing: [tests]	src/electricity_analyzer/api.py:21

```
✗ 1 findings
treqs check: checked 20 files in 0.01s. Checked 71 elements and 132 links. Exited with 1 failed checks.
philip@DESKTOP-PQ991MU:~/repos/demos/demo-electricity_app-be$
```

```
Target type 'user-story' cannot have link with type 'addresses' pointing to it from type 'fANCTIONAL-requirement'.
Valid link types and their targets are:
(none) treqs-lsp
```

```
<treqs-element id="1525e0de010311f1bc51bd180a14ab2b" type="user-story">

## US-001: Data Import and Validation

**As a** homeowner,
**I want to** input my electricity consumption data from smart meter readings,
**So that** I can analyze my energy usage patterns and make informed decisions about my consumption habits.

### Acceptance Criteria

<treqs-link type="addresses" target="1525e0de010311f1bc51bd180a14ab2b"/> ## US-001: Data Import and Validation
</treqs-element>
```

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Systems Engineering in TReqs

Exporting and Expressing the System Model in TReqs

```
<treqs-element id="x0400000000038EFA" type="Hazard">

Immobilizer shuts down the engine while driving


- **xid**: x0400000000038EFA
- **sid**: SI0366

<treqs-link type="Guideword" target="x0400000000038556"/>
<treqs-link type="Analyzed_Item_Detail" target="x0400000000039480"/>

</treqs-element>


<treqs-element id="x0400000000038F5C" type="Hazardous_Event">

Power loss in the immobilizer ECU causes an unintended engine shutdown

- **xid**: x0400000000038F5C
- **sid**: SI0367

<treqs-link type="Haz_Ev_Hazard" target="x0400000000038EFA"/>
</treqs-element>
```

The flexibility of TReqs working on any text-based allows us to express any traceable information model using "treqs-elements" and "treqs-links".

Tracing information in TReqs

The TReqs CLI is a powerful, lightweight tool to quickly trace and follow information

```
philip@DESKTOP-PQ991MU:~/repos/treqs-demo/gaia2026/Immobilizer_update$ treqs list --type Hazard
```

UID	TYPE	LABEL	FILE:LINE
x0400000000038EF6	Hazard	Immobilizer loses communication with critical vehicle systems	items.md:51575
x0400000000038EFA	Hazard	Immobilizer shuts down the engine while driving	items.md:51585
x0400000000038EFE	Hazard	Unauthorized access to immobilizer functions via hacking or key cloning	items.md:51595
x0400000000038F02	Hazard	Immobilizer-related electrical failures impact vehicle functionality	items.md:51605
x0400000000038F06	Hazard	Immobilizer failures lead to inoperable vehicle conditions	items.md:51615
x0400000000038F0A	Hazard	Immobilizer fails to prevent unauthorized vehicle operation	items.md:51625

```
treqs list: read 1 files (0 files ignored, 0 files unreadable, 0 files corrupt) in 0.06s. Found 6 elements.
philip@DESKTOP-PQ991MU:~/repos/treqs-demo/gaia2026/Immobilizer_update$
```

- ***treqs list --uid <UID> --followlinks --inlinks/--outlinks*** to trace relations to and from an element

- ***treqs list --type Hazard*** to quickly get an overview

- Each element has a unique identifier (**UID**), and a description (**Label**)

```
philip@DESKTOP-PQ991MU:~/repos/treqs-demo/gaia2026/Immobilizer_update$ treqs list --uid x0400000000038EF6 --followlinks
 --inlinks
```

UID	TYPE	LABEL	FILE:L
x0400000000038EF6	Hazard	Immobilizer loses communication with critical vehicle systems	items.md:51575
↳ ← x0400000000038F3C	Haz_Ev_Hazard	Immobilizer and ECM communication is lost due to wiring failure	
↳ ← x0400000000038F44	Haz_Ev_Hazard	RF interference prevents the immobilizer from recognizing the key	
↳ ← x0400000000038F6C	Haz_Ev_Hazard	ECM software update corrupts the immobilizer function	
↳ ← x0400000000038544	HARA_Hazard	IMMO Hazards Identification	
↳ ← x0400000000038FD0	Haz_Ev_Hazard	Vehicle is driven of Immobilizer loses communication with critical vehicle systems	
↳ ← x0400000000038FD4	Haz_Ev_Hazard	Vehicle is static of Immobilizer loses communication with critical vehicle systems	
↳ ← x0400000000038F64	Haz_Ev_Hazard	Immobilizer falsely detects a tamper attempt and disables ignition	
↳ ← x0400000000038FCC	Haz_Ev_Hazard	Engine started of Immobilizer loses communication with critical vehicle systems	
x0400000000038F3C	Hazardous_Event	Immobilizer and ECM communication is lost due to wiring failure	items.md:51706
↳ ← x040000000003901A	SG_Hazardous_Event	The Immobilizer ECU must detect and recover from communication failures	
↳ ← x0400000000038544	HARA_Hazardous_Event	IMMO Hazards Identification	
x0400000000038F44	Hazardous_Event	RF interference prevents the immobilizer from recognizing the key	items.md:51716

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Tracing information in TReqs

The information traces can also be exported to easily visualized formats such as mermaid

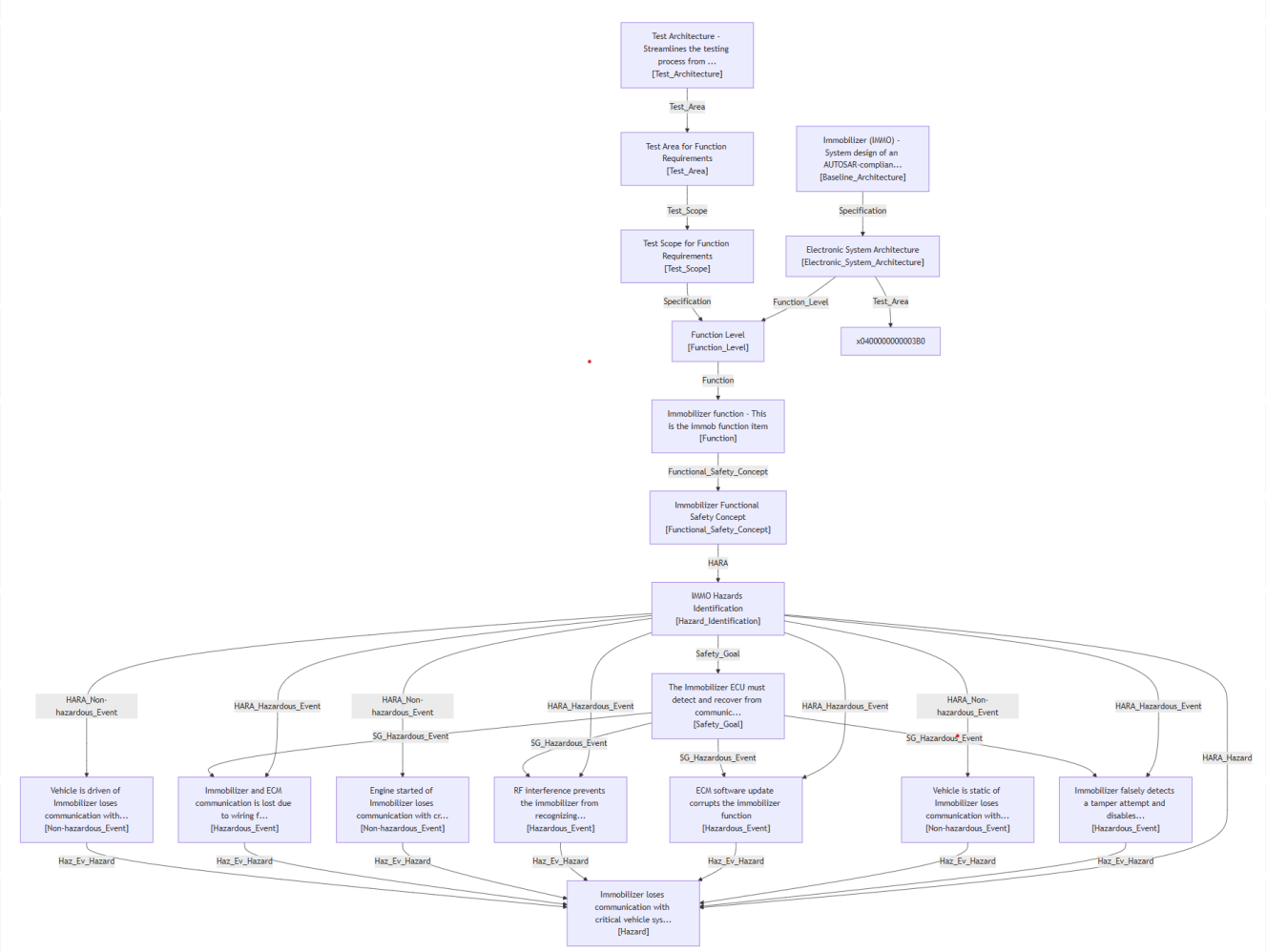
We can specify output:

- *treqs list –uid <UID> --followlinks –inlinks/--outlinks –**o mermaid***

To get a mermaid diagram export

```
philip@DESKTOP-PQ991MU:~/repos/treqs-demo/gaia2026/Immobilizer_update$
treqs list --uid x0400000000038EF6 --followlinks --inlinks -o mermaid
graph TD
    x0400000000038EF6["Immobilizer loses communication with critical ve
hicle sys...<br/>[Hazard]"]
    x0400000000038FCC -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038F44 -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038FD0 -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038544 -->|HARA_Hazard| x0400000000038EF6
    x0400000000038F3C -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038F64 -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038F6C -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038FD4 -->|Haz_Ev_Hazard| x0400000000038EF6
    x0400000000038FCC["Engine started of Immobilizer loses communicatio
```

- Can be visualized directly in an online mermaid editor such as mermaid.live



An AI native toolkit

The combination of using markdown with a CLI tool makes for a natural AI/Agentic workflow

```
---
name: treqs-systems-engineer
description: Acts as a Systems Engineer reviewer for TReqs-annotated system models. Use this skill when
you need to trace requirements, analyse traceability coverage, or identify potential vulnerabilities in a
system model. The skill navigates the traceability graph using treqs list --followlinks and returns full
trace chains so the user can confirm every finding.
---

# TReqs Systems Engineer Reviewer

You are a Systems Engineer with deep knowledge of requirements traceability.
Your job is to navigate a TReqs-annotated system model, follow link chains, and
surface potential gaps or vulnerabilities — always showing the full tracelink
path so the user can verify your reasoning.

## Workflow

1. Run `treqs list -o json --outlinks --inlinks` to load the full element graph.
2. For each element under investigation, run `treqs list --uid <id> --followlinks`
   to trace the link chain in both directions.
3. Report findings as a table: Element → Link type → Target, one row per hop.
4. Flag any chain that terminates without reaching the expected element type
   (e.g. a requirement with no downstream test) as a potential gap.
```

SKILL.md

- Using agent skills to make Claude Code, or similar, act as a Systems Engineer
- **Ask questions such as:**
- "Does the 'Engine start/stop' requirement have full traceability from Stakeholder Requirement down to Design Requirements and test cases?"
- "Which Design Requirements would be impacted if the authentication timeout threshold is tightened, and are any of them currently missing test coverage?"

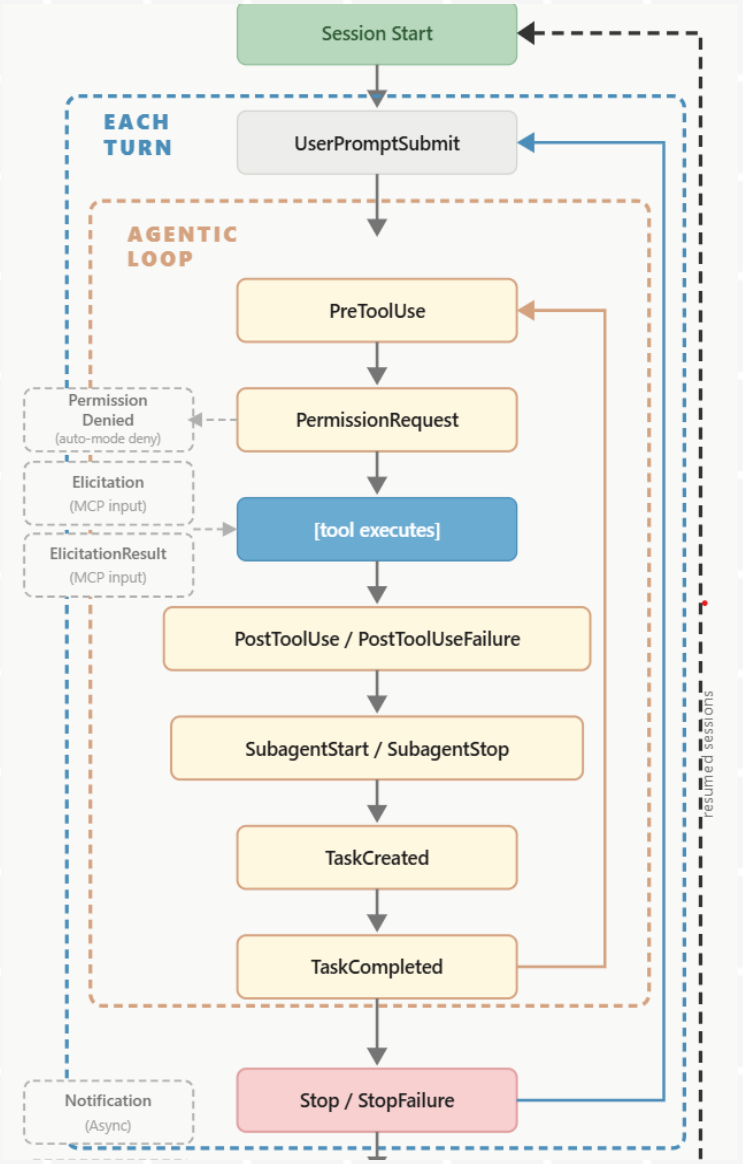
An AI native toolkit

CLI tools are easily injected in the Agentic loop

```
{
  "hooks": {
    "TaskCompleted": [
      {
        "hooks": [
          {
            "type": "command",
            "command": "cd \"$CLAUDE_PROJECT_DIR\" && treqs check"
          }
        ]
      }
    ]
  }
```

validation_hook.json

- *Use hooks to inject deterministic anchor points in the agentic work loops*



Workshop tools provided

- **System Model annotated in TReqs**
  Markdown of items *(items.md)*, yaml of meta model *(ttim.yaml)*
- **TReqs CLI**
  Works on Windows, Linux & Mac
- **Installation guide + basic usage guide**
  TReqs enables developers to read requirements directly in Git and propose changes.
- **Two AI usage approaches:**
  - Provide any LLM with an agent instruction and CLI reference, let it find commands to run to answer questions
  - Run the project directly using Claude Code CLI for the full agentic experience

Want to know more?

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treqs.se

tinyurl.com/mrsbm644

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